Scenario 1 Design Document

Jiahe Ai, Ben Dalton, Rayhan Manoly, Yan Slobodianik

Introduction

Good habits are important for a balanced lifestyle and better mental health. However, building habits is difficult: only 6% of New Year's resolutions last more than a year [1]. A factor contributing to this is that only 20% of people keep themselves accountable [1], meaning that most people require external accountability to help build habits. This is evidenced by a study from the Dominican University of California, which found that "the positive effect of accountability was supported: those who sent weekly progress reports to their friend accomplished significantly more." [2]. An app is needed which allows people to hold each other accountable for building new habits.

To meet this requirement, we will build a community-focused task tracker. Users will join a community with their friends and complete tasks each day to grow a community plant. Through the incentive of growing a plant, users are held accountable by each other to complete their tasks. The target demographic for our app is young adults, as they are more likely to set new year's resolutions [1] (which is often to build a new habit).

Features and Functionality

Having established the need for a social-first approach to task completion, we worked to narrow down the scope of what the app would entail. The purpose of the solution would be to use the idea of a shared target to boost motivation to complete the tasks, as peers within the community would spur each other on.

Building on the success of progress-based apps such as Flora and Duolingo [3,4], we concluded that the shared "vision" within the community would be to grow a plant – a metaphor representative of their own growth. Similar approaches have been successful for people within our target market (teenagers and young adults), as wanting to maintain this progress that everyone has contributed towards proved a sufficient incentive [3].

In our application, the people within the community would choose a common task or habit that they wanted to work on during the week, and the community lead would add the task to the application. Completing the task each day helps keep the plant alive and grow further, whilst also earning currency that can be used to unlock cosmetic upgrades. The reasoning behind rewarding currency is to provide motivation for users. In addition, cosmetic items add customizability to the app, making it more fun to use. The user may also feel that their effort is more meaningful as the

Commented [BD1]: if we need to save space we can reduce the font size from 12 to 11 or 10

Commented [BD2]: i think this should have some mention of advertisements, so the ethics section doesn't seem like its making up potential features for the sake of analysing them

Commented [RM2R2]: added to the bottom of this section

Commented [BD3]: i dont think this sentence is fully justified. what are the similar approaches and what is the evidence of their success?

Commented [RM3R2]: Can cite the reviews on Flora's page. Also, when looking at all their features, there is a lot of overlap, but we differentiate with the shared task approach.

https://flora.appfinca.com/en/

impact on the community is visualised. A collaborative, instead of competitive, approach was taken for the reward mechanism with the intention of disincentivising people from lying about their progress.

To encourage all members of the community to complete the task, the plant's growth depends on the percentage of the community completing the task for the day. Users who have completed the task will be distinguished from those who have not completed the task. This creates social accountability to complete the task, which may give users more motivation compared to if they did not do it as a community.

Currency would be earned individually, based on a user completing their tasks, with bonuses based on completion streaks (e.g. completing a task each day for a week), and on collaborative targets (e.g., everyone in the community completes the task in a day). Refer to Table [2] for further details.

As the users would be using the app to track their progress on the group goals, it would make sense to incorporate a supplementary feature for them to also track their personal habits/to-dos, to keep everything in one place. This feature is useful if the user has personal tasks which they want to keep private or that are not applicable to others. Personal task completions, however, would not be incentivised with progress towards the plant, as it could enable gaming the system.

A more detailed breakdown of the different features and views is shown in Table [1].

Commented [RM4]: Check and number correctly

Commented [RM4R2]:

Core feature/functionality
Home View This displays a fixed number of incomplete tasks with the closest deadlines. The tasks are sorted by their deadlines, where the closer the deadline, the higher up the task is on the list.
Sign In View This view allows the user to sign in and create an account. The view will display if the user has not signed in. The user can see the account creation page by clicking a link on the sign in page. Each account is associated with a number of coins.
Community View In this view, the user can see the daily task in the community. There is also a list of all members. The avatars of those who have completed the task will be highlighted green, whereas those who have not will be highlighted gray. In the center of the view, there is a plant with a level bar on top of it and a growth bar below it. The level bar displays the current level of the plant, and the growth bar displays the daily task completion rate. To improve clarity, the growth bar is colored, where the green section is the range where the plant will level up, the red section is the range where the plant will level down, and the gray section is the number of members who have completed the task. Cosmetic items purchased by members will be displayed on the plant.
Join Community Function

Commented [BD5]: recent deadline means the deadline has already passed

Commented [RM6]: Two health bars, one for daily and one for overall, right?

Commented [JA6R2]: yes

Commented [RM6R3]: can we make this section more concise?

Commented [JA6R4]: I think this is as concise as it can get. To make it shorter perhaps we can move the information in the table into the paragraphs since there's a lot of blank space. Also is "How it address the problem" necessary?

Commented [RM6R5]: Honestly, we could probably avoid it, and add any relevant information to the main text.

A community is associated with a code which a joined user can see. Users that are not in the community can join it by clicking on the "Join" button in community view and inputting the code. One user can only be in one community at a time.
Task Creation Function The creator of the community can create tasks using a button in the community view. The task will be repeated daily until the end of the week and contains a title.
Personal Task Creation In the home view, the user can create personal tasks visible only to themselves. They contain a title and a deadline.
Reward Mechanism The app rewards the user for completing community tasks with coins and growth for the tree. For personal tasks, the user receives no reward. For community tasks, completing the task adds growth to the plant's growth bar and rewards the user with coins, which can be used to buy cosmetic items. Refer to Table [2] for more details.
Shop View In this view, the user can buy cosmetic items for their community using coins.
Community Stats View This displays an overview of community contributions over time, in a week-by-week visualization, as well as all-time statistics. This view can be accessed through the Community View.

Table 1 – Overview of Features

Future Features

To further incentivise communities to make progress and create a greater impact, in a future version, we would like to incorporate an option for users to opt in to advertisements, under which, in return, when a community upgrades their plant, we plant a real tree.

Additionally, a wider range of options when setting the duration of personal and community tasks would be something to implement at a later stage, to give users and communities more flexibility over the tasks they can complete on the app.

We also recognise that a big deterrent to completing the habits that people intend to do is procrastination, increasingly through the use of mobile devices unproductively [5]. It would therefore make sense for our application to use the device's Screen Time data to more effectively prompt/remind the user to complete their tasks before opening apps that are likely to be distracting. However, given the time constraints in the project, this may be a feature that would be introduced in a later version of the application, rather than the prototype.

Lastly, some communication could be helpful within the app, given its community-oriented nature. A full-on chat interface would come with a whole host of challenges, especially from an encryption/storage standpoint, proving infeasible, though in some future version of the app, a chat option with limited options for sending messages (i.e. to prompt friends to complete their task, or to choose next week's activity), could be built in.

Frameworks & Languages

Pivoting now to our technical details, we decided to revolve our application around the Expo framework. It's based on React Native, which provides us the ability to use a single codebase and language (React with TypeScript) to support both iOS and Android mobile platforms, instead of having to use two, arguably harder languages. The primitive building blocks that Expo ships are perfect for maintaining consistency and a quick iteration cycle.

We can split our choice of tooling into two parts: frontend and backend. A technical breakdown of the different elements needed to develop the features for this application is detailed below.

Due to Expo already supplying us with the basic primitives for building system-consistent interfaces, our frontend will not need much more besides Nativewind, a styling solution based on the popular web-based framework TailwindCSS, providing us with the simplest approach to view styling, resulting in the fastest design iteration cycle and maintainability. In terms of our backend, we landed on a relatively new tool called Convex.

Convex replaces many tools we would otherwise require, like a hosted database provider, end-to-end type checkers like tRPC, compute runner for providing an API, and websockets for real-time synchronization with a single tool. Convex works perfectly for us as it provides amazing tooling for solving challenges our app would face. For instance, it is critical that our app remains in sync across our users' devices (Convex is real-time) [9], performant due to a high amount of database read queries expected (Convex queries are heavily optimised and cacheable), and cheap (one SaaS product versus many).

However, Convex does not solve all our issues, a major one being authentication. For this, we will use a hosted service called Clerk. We chose Clerk mostly due to its great support on Expo, being a hosted option (the responsibility of uptime of the authentication service doesn't fall on us) and due to its pricing model. We foresee many users signing up to our app once to try it out without ever returning to use it. Fortunately, Clerk uniquely does NOT count one-off sign-ins towards its 50,000 free Monthly Retained Users [10], which will long-term save us on expenses.

Another library we will use, called react-native-device-activity, will provide us with "direct access to Apple's Screen Time, Device Activity, and Shielding APIs" [11]. This will allow us to integrate the screen-time and app activity monitoring features described in the section above for iOS devices. One caveat to using these APIs is the requirement for distributed apps to gain Family Controls entitlement from Apple. This means that after testing the features in a development environment, for the app to be able to be published with these features working, it needs to be distributed by someone in the Apple Developer Program and have these entitlements. For that reason, depending on feasibility during development, we may decide this is best left to a later stage.

Commented [BD7]: screen time is not yet written in the section above

Commented [RM7R2]: have added that section now.

Commented [RM8]: Is this line okay?

Ethics Analysis

Social Accountability

Being community-centric rather than independent, our app is more likely to help people build new habits, as friends hold each other accountable [2]. This is important ethically as it means we are being beneficent; we hold the best interests (building habits) of our users as our top priority.

However, social accountability is a double-edged sword. While it can lead to accomplishment, it can also lead to peer pressure. Peer pressure has been linked to negative mental health, depression, stress, and anxiety [6]. To follow the Engineering Council (EC) and Royal Academy of Engineering (RAE) principle to "hold paramount the health and safety of others" [7], we must minimise the possibility of peer pressure occurring on the app. Given that our aim is to have friend groups sign up together, we cannot fully mitigate the possibility of peer pressure occurring outside the app. To protect our users, we will not implement any sources of communication directly on the app, and we can implement a disclaimer about the effects of peer pressure when a user creates an account. Another solution is to grant leniency in growing the plant; we will require only a certain percentage of the community to complete their daily tasks to grow the plant.

Our current design is a trust-based system in which community members are not required to verify completion of an activity. Users may feel pressured to tick off a task without completing it to grow the community plant. This should be avoided so that our users can get the most out of our app and build new habits. In a future iteration, we could add BeReal-inspired photo verification so that other members of the community have evidence of who has completed their task.

Confidentiality

The EC and RAE recognise the "importance of physical and cyber security and data protection" [7], along with the confidentiality of others [7]. This applies to our app because we will store sensitive data, such as passwords and what habits our users want to build. By implementing rigorous data validation on our back end, we can protect our database from attacks such as SQL injections and keep the data secure. Furthermore, we can hash user passwords, which makes it impossible to log in as another user even if the database is compromised.

Monetization

If a community grows their plant for an arbitrary number of consecutive weeks, we can reward them by donating to #teamtrees, tying the abstract progress on the app to a concrete goal. This will encourage them to complete their goals. The capital required for the donation can be raised through advertisements (ads). The ethics of ads must be considered, while we are following the EC and RAE principle to "protect, and where possible improve" our environment [7], ads have long been used to manipulate the people they are targeted towards [8]. To protect our users from manipulation, we can make the ads an opt-in scheme where, on joining, the users are warned of their danger. Furthermore, we can ensure that the ad provider has a reputation for ethical practices.

Commented [RM9]: @Ben Dalton could you add something here about the "trust" element of completing a task, and that long term, perhaps we could have a BeReal-style "snap a photo during your task" form of "verification"?

Commented [BD9R2]: done - see third paragraph under social accountability section

Commented [RM10]: change phrasing to sound less essential, else the question arises as to why we don't implement it now

Commented [BD10R2]: must -> should, i think thats ok now

Technical Design Details

Database Structure

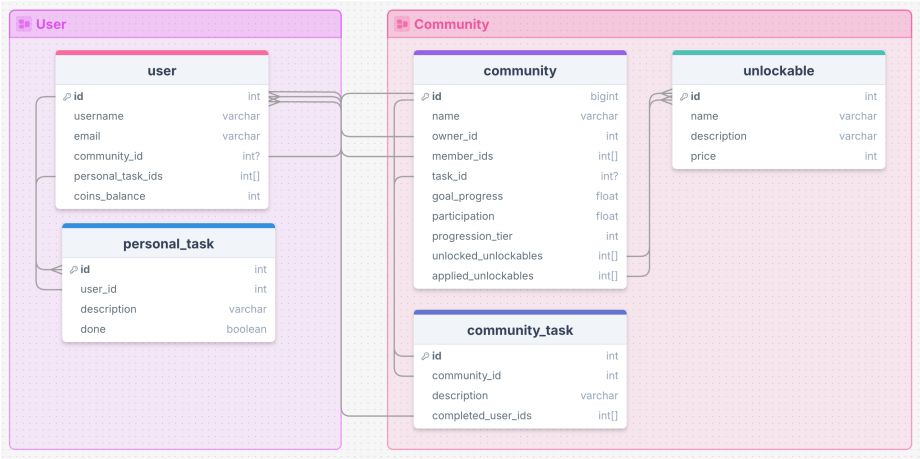

Figure 1 - Database Structure Diagram

Our database structure will be split into two overall sections, one pertaining to the user, and the other to the community. When a user downloads our app and goes through the onboarding process, a new entry is added to the user table. This will store our users' information, along with any personal tasks that they create, which will be linked to them for easy access. If a user decides to create a community, a new row in the communities table will be created, with the shape that you see above. It will store all the info about the community, along with any members and, importantly, the user ID of the user who created the community. This user will be considered the community "owner" and will be the only one with permission to kick members and create community tasks. These community tasks will be stored in their own table, along with a list of member IDs who completed them. Using the size of that array combined with the size of the corresponding community's member list is how our app will be able to calculate the proportion of members who completed the community goal to award participation incentives (see below). With these "coins", community members will be able to purchase unlockable cosmetics (represented by the "unlockable" table in Fig. 1) to visually see the progress in their goals. The communities will also store their progress in our app's progression system, another way to visualise the community's long-term success in their tasks.

Scoring Process

To incentivise continuous progress, we decided that a threshold-based system would be optimal. That is, a certain task completion rate is needed to sustain the plant in a day, and each member of the community can contribute by completing their tasks. The plant's level would also be displayed, with “better” plants being unlocked at certain levels.

The plant’s growth depends on the percentage of the community that completes the task; less than 50% causes the plant to wither, decreasing its level by 1, whereas greater than 50% causes it to grow the next day, according to the difference from 50% (e.g. if 50% is 4 people, and 6 people complete the task in the day, the plant grows +2), and making progress towards the next plant. When the plant reaches the base level of the next type of plant, it upgrades to the next plant. Additionally, coins can be earned for cosmetic upgrades to the plant.

Commented [JA11]: Is using < in a sentence appropriate?

Commented [JA11R2]: made some adjustments such as changing token to completion rate and making level up mechanism clearer and more explicit

Coins are earned according to the details in Table [2].

Description of Task	Individual or Collective?	Number of Coins Earned
Completing your daily task	Individual	+1
Completing your daily task for the whole week	Individual	+3
Entire community completing the task in a given day	Collective	+2

Table 2 – Breakdown of Coin Distribution

The structure of the reward system is designed to incentivise both consistency and regular participation of the entire community for the best rewards.

Assets required

As this application is gamified somewhat, there are a number of assets that would make the progression of the plant more immersive. Primarily, we'd need assets to represent the different progressions of plants that the community can unlock. For the scope of the prototype, we found three would be sufficient, though later on, more could be added to ensure there's sufficient development as communities progress.

Additionally, assets for cosmetic upgrades to the plants (e.g. pots, colours), would also need development, and, as with the plants themselves, the prototype will have a limited collection to be expanded on at a later point.

The primary view, the community view, would be laid out in a manner similar to the wireframe in Figure [2], containing all the essential information to determine the plant's progress, both towards the next level (as indicated by the progress bar above the plant), and for the day (as indicated by the lower progress bar). Other pages, such as the individual tasks and shop page, are more standard in their layout.

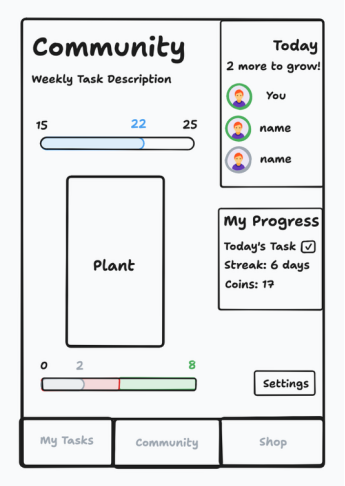

Figure 3 - Wireframe of the Community Page

Bibliography

1. Forbes, “New Year’s Resolutions Statistics” [Online]. Available: https://www.forbes.com/health/mind/new-years-resolutions-statistics/ [Accessed 9 Feb 2026]
2. G. Matthews, “The Impact of Commitment, Accountability, and Written Goals on Goal Achievement” [Online]. Available: https://scholar.dominican.edu/psychology-faculty-conference-presentations/3/ [Accessed 9 Feb 2026]
3. AppFinca. Inc, “Flora - Green Focus,” [Online]. Available: https://flora.appfinca.com/en/ [Accessed 12 Feb 2026]
4. O. Mansur, “The habit-building research behind your Duolingo streak,” *Duolingo Blog*, Jan. 31, 2022. Available: https://blog.duolingo.com/how-duolingo-streak-builds-habit/ [Accessed 12 Feb 2026]
5. A. M. Roffarello and L. De Russis, “Understanding, Discovering, and Mitigating Habitual Smartphone Use in Young Adults,” ACM Transactions on Interactive Intelligent Systems, vol. 11, no. 2, pp. 1–34, Jul. 2021, doi: https://doi.org/10.1145/3447991.
6. Mentally Healthy Schools, “Peer Pressure” [Online]. Available: http://mentallyhealthyschools.org.uk/factors-that-impact-mental-health/school-based-risk-factors/peer-pressure/ [Accessed 8 Feb 2026]
7. Engineering Council, Royal Academy of Engineering, “Statement of Ethical Principles” [Online]. Available: https://www.engc.org.uk/media/hjjdmlet/guidance-on-ethical-principles-leaflet.pdf [Accessed 8 Feb 2026]
8. M. Arango-Kure, M. Garz, “Manipulation: An integrative framework of unethical influence in marketing” [Online]. Available: https://doi.org/10.1016/j.jbusres.2025.115476 [Accessed 8 Feb 2026]
9. Convex, Inc, “Realtime, all the time”, Convex, [Online]. Available: https://www.convex.dev/realtime [Accessed 13 February 2026]
10. Clerk, Inc, "Pricing," Clerk, [Online]. Available: https://clerk.com/pricing [Accessed 10 February 2026].
11. kingstinct, "Kingstinct/react-native-device-activity," GitHub, [Online]. Available: https://github.com/Kingstinct/react-native-device-activity [Accessed 10 February 2026].